

Importing Libraries

```
In [4]: # Data Manipulation
import pandas as pd
import numpy as np

# Visualization
import matplotlib.pyplot as plt
import seaborn as sns

# Ignore warnings
import warnings
warnings.filterwarnings("ignore")

# Better plot style
sns.set_style("whitegrid")

print("Libraries imported successfully!")
```

Libraries imported successfully!

Loading the dataset

```
In [11]: df = pd.read_csv(r"C:\Users\ali\OneDrive\Desktop\projects\pro_env\Hotel_Analysis\Da
```

Exploratory Data Analysis and Data cleaning

```
In [14]: df.head() # return top 5 rows
df.sample(5)
```

Out[14]:

	hotel	is_canceled	lead_time	arrival_date_year	arrival_date_month	arrival_date_we
75089	City Hotel	1	283	2015	July	
95886	City Hotel	0	7	2016	August	
74719	City Hotel	0	0	2016	May	
47020	City Hotel	0	58	2016	February	
36632	Resort Hotel	0	7	2017	May	

5 rows × 36 columns

In [15]: `df.shape`

Out[15]: (119390, 36)

In [16]: `df.columns`

Out[16]: Index(['hotel', 'is_canceled', 'lead_time', 'arrival_date_year',
 'arrival_date_month', 'arrival_date_week_number',
 'arrival_date_day_of_month', 'stays_in_weekend_nights',
 'stays_in_week_nights', 'adults', 'children', 'babies', 'meal',
 'country', 'market_segment', 'distribution_channel',
 'is_repeated_guest', 'previous_cancellations',
 'previous_bookings_not_canceled', 'reserved_room_type',
 'assigned_room_type', 'booking_changes', 'deposit_type', 'agent',
 'company', 'days_in_waiting_list', 'customer_type', 'adr',
 'required_car_parking_spaces', 'total_of_special_requests',
 'reservation_status', 'reservation_status_date', 'name', 'email',
 'phone-number', 'credit_card'],
 dtype='str')

In [18]: `df.info(all)`

```

<class 'pandas.DataFrame'>
RangeIndex: 119390 entries, 0 to 119389
Data columns (total 36 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   hotel                                119390 non-null  str
1   is_canceled                          119390 non-null  int64
2   lead_time                            119390 non-null  int64
3   arrival_date_year                    119390 non-null  int64
4   arrival_date_month                   119390 non-null  str
5   arrival_date_week_number             119390 non-null  int64
6   arrival_date_day_of_month             119390 non-null  int64
7   stays_in_weekend_nights               119390 non-null  int64
8   stays_in_week_nights                  119390 non-null  int64
9   adults                                119390 non-null  int64
10  children                              119386 non-null  float64
11  babies                                119390 non-null  int64
12  meal                                  119390 non-null  str
13  country                              118902 non-null  str
14  market_segment                       119390 non-null  str
15  distribution_channel                  119390 non-null  str
16  is_repeated_guest                     119390 non-null  int64
17  previous_cancellations                 119390 non-null  int64
18  previous_bookings_not_canceled         119390 non-null  int64
19  reserved_room_type                    119390 non-null  str
20  assigned_room_type                     119390 non-null  str
21  booking_changes                       119390 non-null  int64
22  deposit_type                           119390 non-null  str
23  agent                                 103050 non-null  float64
24  company                               6797 non-null   float64
25  days_in_waiting_list                  119390 non-null  int64
26  customer_type                         119390 non-null  str
27  adr                                   119390 non-null  float64
28  required_car_parking_spaces            119390 non-null  int64
29  total_of_special_requests              119390 non-null  int64
30  reservation_status                    119390 non-null  str
31  reservation_status_date                119390 non-null  str
32  name                                  119390 non-null  str
33  email                                  119390 non-null  str
34  phone-number                           119390 non-null  str
35  credit_card                           119390 non-null  str
dtypes: float64(4), int64(16), str(16)
memory usage: 32.8 MB

```

```

In [22]: pd.set_option("display.width", None)
         pd.set_option("display.max_columns", None)

```

```

In [23]: df.info()

```

```

<class 'pandas.DataFrame'>
RangeIndex: 119390 entries, 0 to 119389
Data columns (total 36 columns):
 #   Column                                Non-Null Count  Dtype
---  -
 0   hotel                                119390 non-null  str
 1   is_canceled                          119390 non-null  int64
 2   lead_time                            119390 non-null  int64
 3   arrival_date_year                    119390 non-null  int64
 4   arrival_date_month                  119390 non-null  str
 5   arrival_date_week_number             119390 non-null  int64
 6   arrival_date_day_of_month            119390 non-null  int64
 7   stays_in_weekend_nights              119390 non-null  int64
 8   stays_in_week_nights                 119390 non-null  int64
 9   adults                               119390 non-null  int64
10  children                             119386 non-null  float64
11  babies                               119390 non-null  int64
12  meal                                 119390 non-null  str
13  country                             118902 non-null  str
14  market_segment                       119390 non-null  str
15  distribution_channel                  119390 non-null  str
16  is_repeated_guest                     119390 non-null  int64
17  previous_cancellations                 119390 non-null  int64
18  previous_bookings_not_canceled         119390 non-null  int64
19  reserved_room_type                    119390 non-null  str
20  assigned_room_type                    119390 non-null  str
21  booking_changes                       119390 non-null  int64
22  deposit_type                          119390 non-null  str
23  agent                                 103050 non-null  float64
24  company                              6797 non-null   float64
25  days_in_waiting_list                  119390 non-null  int64
26  customer_type                         119390 non-null  str
27  adr                                   119390 non-null  float64
28  required_car_parking_spaces           119390 non-null  int64
29  total_of_special_requests              119390 non-null  int64
30  reservation_status                    119390 non-null  str
31  reservation_status_date                119390 non-null  str
32  name                                  119390 non-null  str
33  email                                 119390 non-null  str
34  phone-number                          119390 non-null  str
35  credit_card                           119390 non-null  str
dtypes: float64(4), int64(16), str(16)
memory usage: 32.8 MB

```

```

In [25]: # reservation_status_date str to date time series

df['reservation_status_date'] = pd.to_datetime(df['reservation_status_date'])

print(df['reservation_status_date'].dtype)

datetime64[us]

```

```

In [26]: # describe function use for numerical but I use for cateforical with include parame

df.describe(include="object")

```

Out[26]:

	hotel	arrival_date_month	meal	country	market_segment	distribution_channel
count	119390	119390	119390	118902	119390	119390
unique	2	12	5	177	8	5
top	City Hotel	August	BB	PRT	Online TA	TA/TC
freq	79330	13877	92310	48590	56477	97870

```
In [33]: for col in df.describe(include="object").columns:
          print("="*50)
          print(f"column name: {col}")
          print(f"unique value: {df[col].nunique()}")
          print("-"*50)
          print(df[col].unique(), "\n")
```

```
=====
column name: hotel
unique value: 2
-----
<StringArray>
['Resort Hotel', 'City Hotel']
Length: 2, dtype: str

=====
column name: arrival_date_month
unique value: 12
-----
<StringArray>
[      'July',      'August', 'September',  'October',  'November',  'December',
      'January', 'February',   'March',    'April',    'May',      'June']
Length: 12, dtype: str

=====
column name: meal
unique value: 5
-----
<StringArray>
['BB', 'FB', 'HB', 'SC', 'Undefined']
Length: 5, dtype: str

=====
column name: country
unique value: 177
-----
<StringArray>
['PRT', 'GBR', 'USA', 'ESP', 'IRL', 'FRA',   nan, 'ROU', 'NOR', 'OMN',
 ...
 'ATA', 'GTM', 'ASM', 'MRT', 'NCL', 'KIR', 'SDN', 'ATF', 'SLE', 'LAO']
Length: 178, dtype: str

=====
column name: market_segment
unique value: 8
-----
<StringArray>
[      'Direct',      'Corporate',      'Online TA', 'Offline TA/TO',
      'Complementary',      'Groups',      'Undefined',      'Aviation']
Length: 8, dtype: str

=====
column name: distribution_channel
unique value: 5
-----
<StringArray>
['Direct', 'Corporate', 'TA/TO', 'Undefined', 'GDS']
Length: 5, dtype: str

=====
column name: reserved_room_type
unique value: 10
-----
```

```
<StringArray>
['C', 'A', 'D', 'E', 'G', 'F', 'H', 'L', 'P', 'B']
Length: 10, dtype: str
```

```
=====
column name: assigned_room_type
unique value: 12
-----
```

```
<StringArray>
['C', 'A', 'D', 'E', 'G', 'F', 'I', 'B', 'H', 'P', 'L', 'K']
Length: 12, dtype: str
```

```
=====
column name: deposit_type
unique value: 3
-----
```

```
<StringArray>
['No Deposit', 'Refundable', 'Non Refund']
Length: 3, dtype: str
```

```
=====
column name: customer_type
unique value: 4
-----
```

```
<StringArray>
['Transient', 'Contract', 'Transient-Party', 'Group']
Length: 4, dtype: str
```

```
=====
column name: reservation_status
unique value: 3
-----
```

```
<StringArray>
['Check-Out', 'Canceled', 'No-Show']
Length: 3, dtype: str
```

```
=====
column name: name
unique value: 81503
-----
```

```
<StringArray>
[      'Ernest Barnes',      'Andrea Baker',      'Rebecca Parker',
        'Laura Murray',      'Linda Hines',      'Jasmine Fletcher',
        'Dylan Rangel',      'William Velez',      'Steven Murphy',
        'Michael Moore',
        ...
        'Mary Pittman',      'Meghan Edwards',      'Martha Acevedo',
        'Betty Taylor',      'Cesar Moreno',      'Jose Andersen',
        'James Mccarty',      'Wesley Aguilar',      'Caroline Conley MD',
        'Ariana Michael']
Length: 81503, dtype: str
```

```
=====
column name: email
unique value: 115889
-----
```

```
<StringArray>
[ 'Ernest.Barnes31@outlook.com',      'Andrea_Baker94@aol.com',
  'Rebecca_Parker@comcast.net',      'Laura_M@gmail.com',
    'LHines@verizon.com',            'JFletcher43@xfinity.com',
  'Rangel.Dylan@comcast.net',        'Velez_William@mail.com',
  'Steven.Murphy54@aol.com',         'MichaelMoore81@outlook.com',
  ...
    'Wanda_J@yandex.com',            'BTaylor@yandex.com',
  'Cesar_M50@comcast.net',           'Andersen_Jose@protonmail.com',
    'James_M@att.com',               'Claudia.J@yahoo.com',
  'WAguilar@xfinity.com',            'Mary_Morales@hotmail.com',
  'MD_Caroline@comcast.net',         'Ariana_M@xfinity.com']
Length: 115889, dtype: str
```

```
=====
column name: phone-number
unique value: 119390
-----
```

```
<StringArray>
['669-792-1661', '858-637-6955', '652-885-2745', '364-656-8427',
 '713-226-5883', '190-271-6743', '420-332-5209', '286-669-4333',
 '341-726-5787', '316-648-6176',
 ...
 '379-374-7947', '114-632-1395', '666-379-7260', '656-714-5930',
 '689-204-7523', '403-092-5582', '238-763-0612', '395-518-4100',
 '531-528-1017', '422-804-6403']
Length: 119390, dtype: str
```

```
=====
column name: credit_card
unique value: 9000
-----
```

```
<StringArray>
['*****4322', '*****9157', '*****3734',
 '*****5677', '*****5498', '*****9263',
 '*****6994', '*****8729', '*****3639',
 '*****9190',
 ...
 '*****1483', '*****4720', '*****1966',
 '*****4212', '*****2986', '*****1734',
 '*****5780', '*****9170', '*****6349',
 '*****7959']
Length: 9000, dtype: str
```

```
In [34]: df.isnull().sum()
```



```

Out[34]: hotel                                0
         is_canceled                          0
         lead_time                           0
         arrival_date_year                   0
         arrival_date_month                 0
         arrival_date_week_number           0
         arrival_date_day_of_month          0
         stays_in_weekend_nights            0
         stays_in_week_nights               0
         adults                              0
         children                            4
         babies                             0
         meal                               0
         country                            488
         market_segment                     0
         distribution_channel                0
         is_repeated_guest                  0
         previous_cancellations              0
         previous_bookings_not_canceled     0
         reserved_room_type                 0
         assigned_room_type                 0
         booking_changes                     0
         deposit_type                       0
         agent                             16340
         company                             112593
         days_in_waiting_list                0
         customer_type                      0
         adr                                0
         required_car_parking_spaces         0
         total_of_special_requests           0
         reservation_status                 0
         reservation_status_date             0
         name                               0
         email                              0
         phone-number                       0
         credit_card                        0
         dtype: int64

```

```

In [42]: # check in percentages

company_null_per = round((df['company'].isnull().sum()/ len(df)) * 100,2)
print(f"null percentag: {company_null_per}%")

```

null percentag: 94.31%

```

In [49]: for col, val in zip(df.columns, (round((df.isnull().mean() * 100), 2))):
         print(f"{col} : {val}")

```

```

hotel : 0.0
is_canceled : 0.0
lead_time : 0.0
arrival_date_year : 0.0
arrival_date_month : 0.0
arrival_date_week_number : 0.0
arrival_date_day_of_month : 0.0
stays_in_weekend_nights : 0.0
stays_in_week_nights : 0.0
adults : 0.0
children : 0.0
babies : 0.0
meal : 0.0
country : 0.41
market_segment : 0.0
distribution_channel : 0.0
is_repeated_guest : 0.0
previous_cancellations : 0.0
previous_bookings_not_canceled : 0.0
reserved_room_type : 0.0
assigned_room_type : 0.0
booking_changes : 0.0
deposit_type : 0.0
agent : 13.69
company : 94.31
days_in_waiting_list : 0.0
customer_type : 0.0
adr : 0.0
required_car_parking_spaces : 0.0
total_of_special_requests : 0.0
reservation_status : 0.0
reservation_status_date : 0.0
name : 0.0
email : 0.0
phone-number : 0.0
credit_card : 0.0

```

```

In [51]: missing_data = pd.DataFrame({
        "Missing Count": df[["country", "agent", "company"]].isnull().sum(),
        "Missing %": round(df[["country", "agent", "company"]].isnull().mean() * 100, 2
    })

missing_data

```

```
Out[51]:
```

	Missing Count	Missing %
country	488	0.41
agent	16340	13.69
company	112593	94.31

```

In [53]: # Now drop company, agent --> why? --> because company have too much null and agent

```

```
df.drop(['company', 'agent'], axis=1, inplace=True)
df.dropna(inplace=True) # country null rows deleted, no impact on my entire dataset.
```

```
In [54]: df.isnull().sum()
```

```
Out[54]: hotel                                0
is_canceled                                0
lead_time                                  0
arrival_date_year                          0
arrival_date_month                         0
arrival_date_week_number                   0
arrival_date_day_of_month                  0
stays_in_weekend_nights                    0
stays_in_week_nights                      0
adults                                    0
children                                  0
babies                                    0
meal                                       0
country                                  0
market_segment                            0
distribution_channel                       0
is_repeated_guest                         0
previous_cancellations                     0
previous_bookings_not_canceled             0
reserved_room_type                         0
assigned_room_type                         0
booking_changes                           0
deposit_type                              0
days_in_waiting_list                     0
customer_type                             0
adr                                         0
required_car_parking_spaces               0
total_of_special_requests                 0
reservation_status                         0
reservation_status_date                   0
name                                       0
email                                      0
phone-number                             0
credit_card                               0
dtype: int64
```

```
In [61]: df.describe() # only numerical values summaries
```

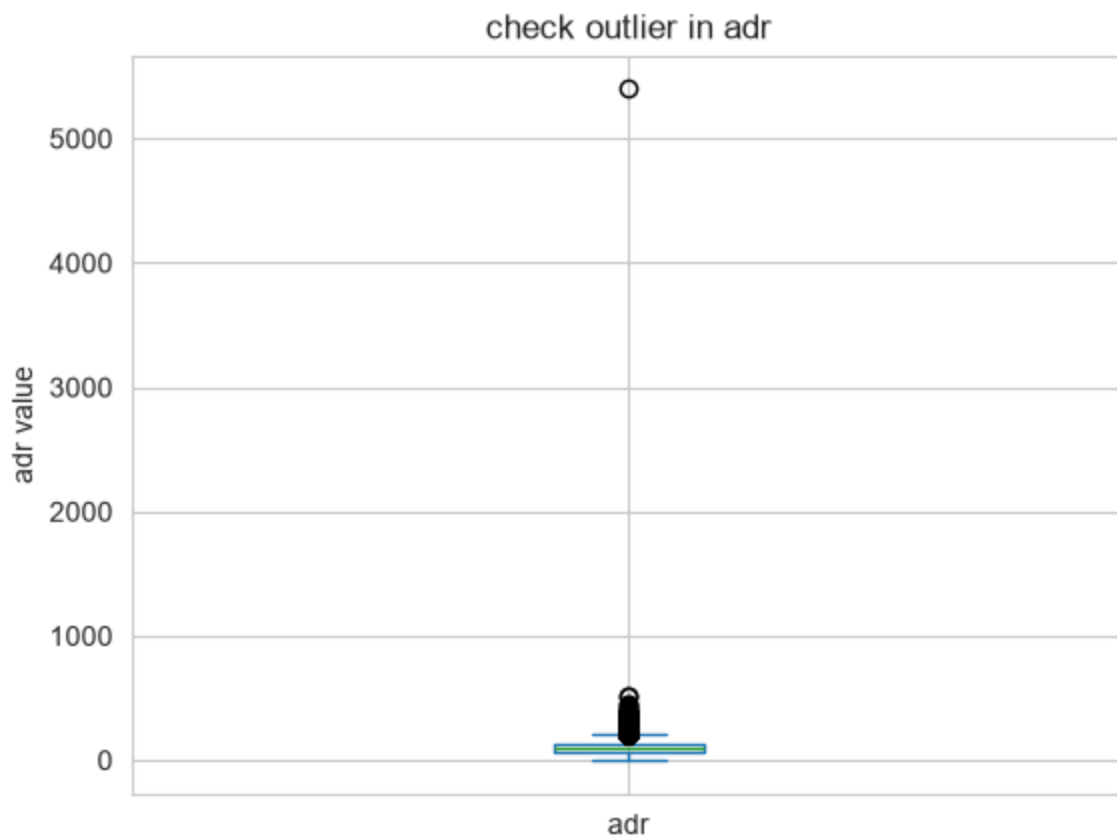
Out[61]:

	is_canceled	lead_time	arrival_date_year	arrival_date_week_number	arrival_date
count	118898.000000	118898.000000	118898.000000	118898.000000	
mean	0.371352	104.311435	2016.157656	27.166555	
min	0.000000	0.000000	2015.000000	1.000000	
25%	0.000000	18.000000	2016.000000	16.000000	
50%	0.000000	69.000000	2016.000000	28.000000	
75%	1.000000	161.000000	2017.000000	38.000000	
max	1.000000	737.000000	2017.000000	53.000000	
std	0.483168	106.903309	0.707459	13.589971	

In [63]: *# adr has outlier*

```
df['adr'].plot(kind='box')
plt.title("check outlier in adr")
plt.ylabel("adr value")

plt.show()
```

In [64]: *# remove outlier only in adr*

```
df = df[df['adr'] < 5000]
```

Data Analysis and Visualization

```
In [68]: cancelled_perc = df['is_canceled'].value_counts(normalize=True) * 100
print(cancelled_perc)
```

```
is_canceled
0    62.865337
1    37.134663
Name: proportion, dtype: float64
```

```
In [70]: counts = df['is_canceled'].value_counts()

plt.figure(figsize=(6,4))

plt.bar(
    ['Not Canceled', 'Canceled'],
    counts,
    color=['green', 'red'],          # Bar colors
    edgecolor='black',              # Border color
    linewidth=2,                    # Border thickness
    width=0.5,                      # Bar width
    label=f"Not Canceled: {cancelled_perc[0]:.2f}%\nCanceled: {cancelled_perc[1]:.2f}%"
)

plt.xlabel("Booking Status")
plt.ylabel("Number of Bookings")
plt.title("Hotel Booking Cancellation")
plt.legend()

plt.show()
```



```
In [77]: plt.figure(figsize=(8,4))

ax1 = sns.countplot(x= 'hotel', hue = 'is_canceled', data=df, palette='Blues')

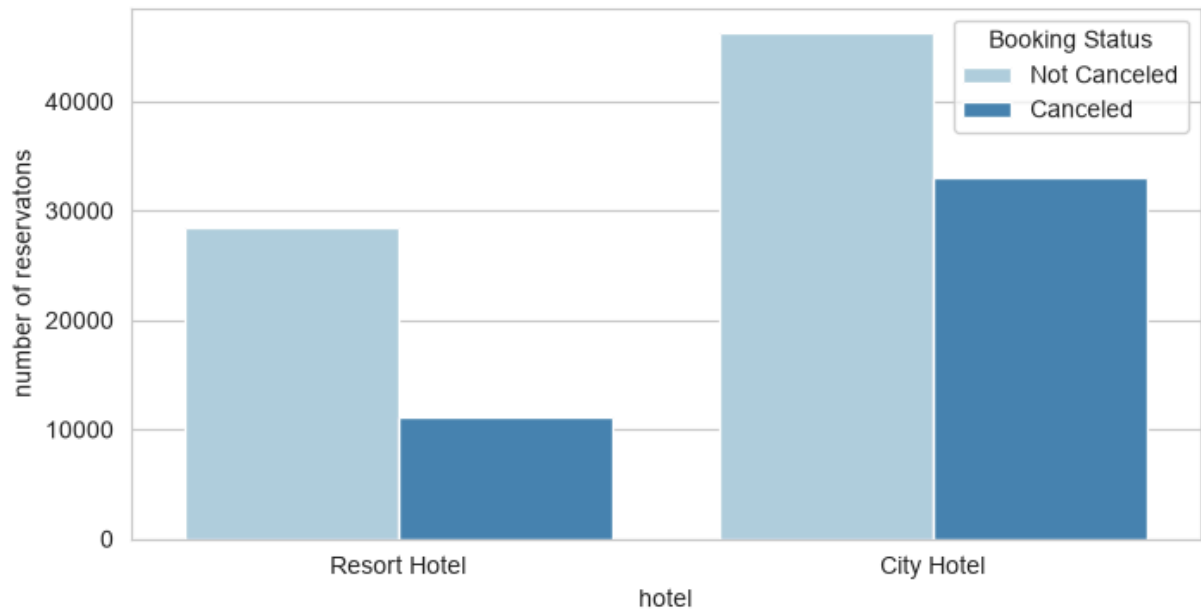
handles, labels = ax1.get_legend_handles_labels()

ax1.legend(
    handles,
    ['Not Canceled', 'Canceled'],
    title='Booking Status',
    bbox_to_anchor=(1,1)
)

plt.title("Reservation status i different hotels", size = 20)
plt.xlabel("hotel")
plt.ylabel("number of reservatons")
```

Out[77]: Text(0, 0.5, 'number of reservatons')

Reservation status i different hotels



```
In [78]: resort_hotel = df[df['hotel'] == 'Resort Hotel']
resort_hotel['is_canceled'].value_counts(normalize=True)
```

```
Out[78]: is_canceled
0    0.72025
1    0.27975
Name: proportion, dtype: float64
```

```
In [79]: city_hotel = df[df['hotel'] == 'City Hotel']
city_hotel['is_canceled'].value_counts(normalize=True)
```

```
Out[79]: is_canceled
0    0.582918
1    0.417082
Name: proportion, dtype: float64
```

```
In [80]: resort_hotel = resort_hotel.groupby('reservation_status_date')[['adr']].mean()
city_hotel = city_hotel.groupby('reservation_status_date')[['adr']].mean()
```

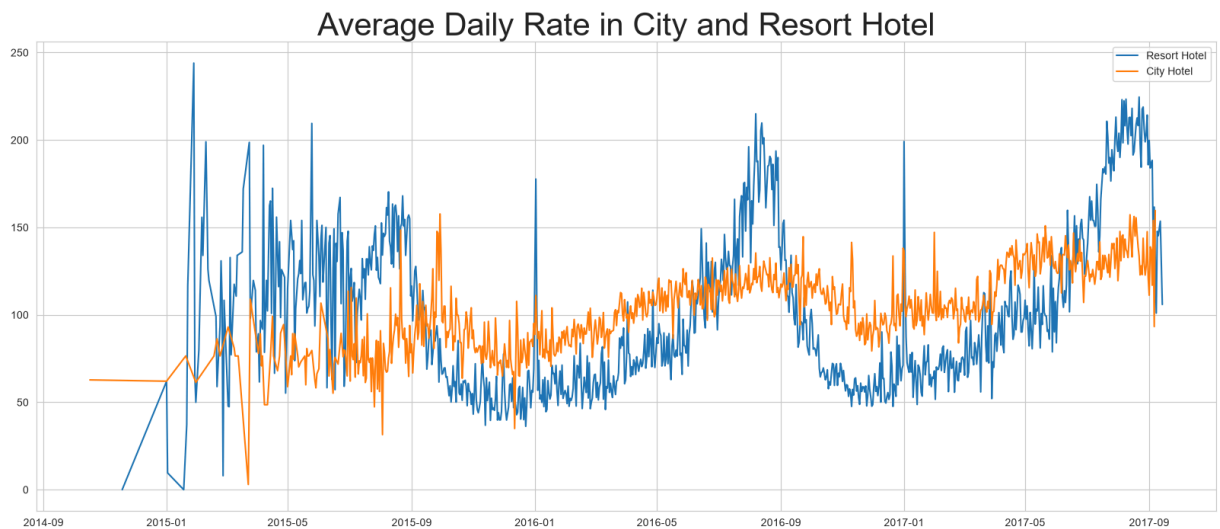
```
In [85]: plt.figure(figsize=(20,8))

plt.title('Average Daily Rate in City and Resort Hotel', fontsize=30)

plt.plot(
    resort_hotel.index,
    resort_hotel['adr'],
    label='Resort Hotel'
)

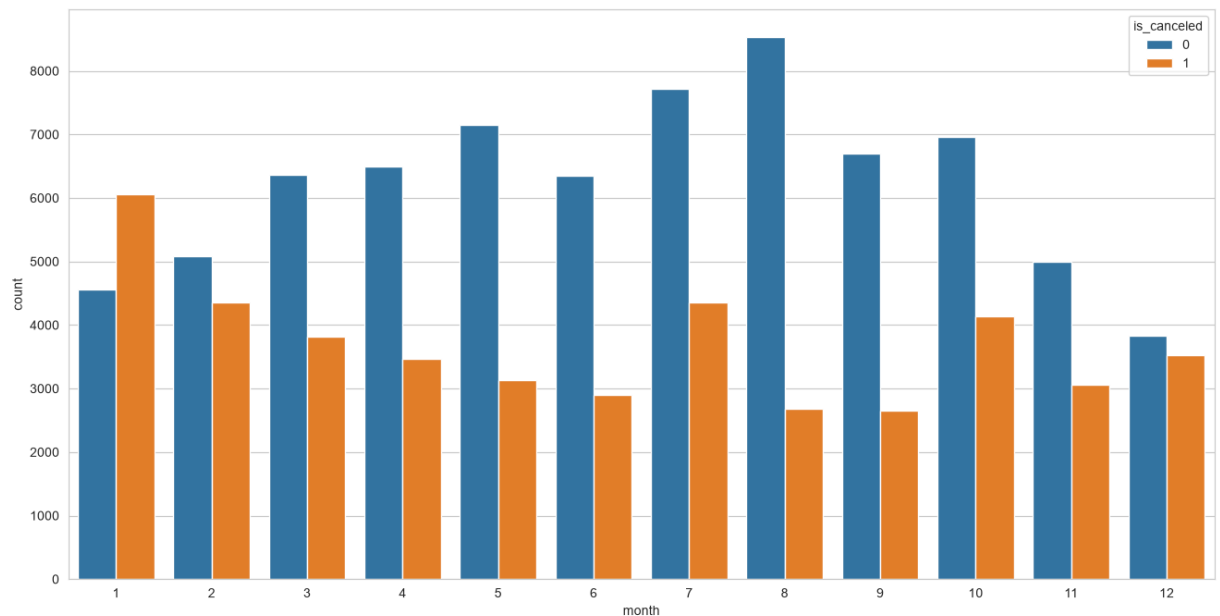
plt.plot(
    city_hotel.index,
    city_hotel['adr'],
    label='City Hotel'
)
```

```
plt.legend()
plt.show()
```

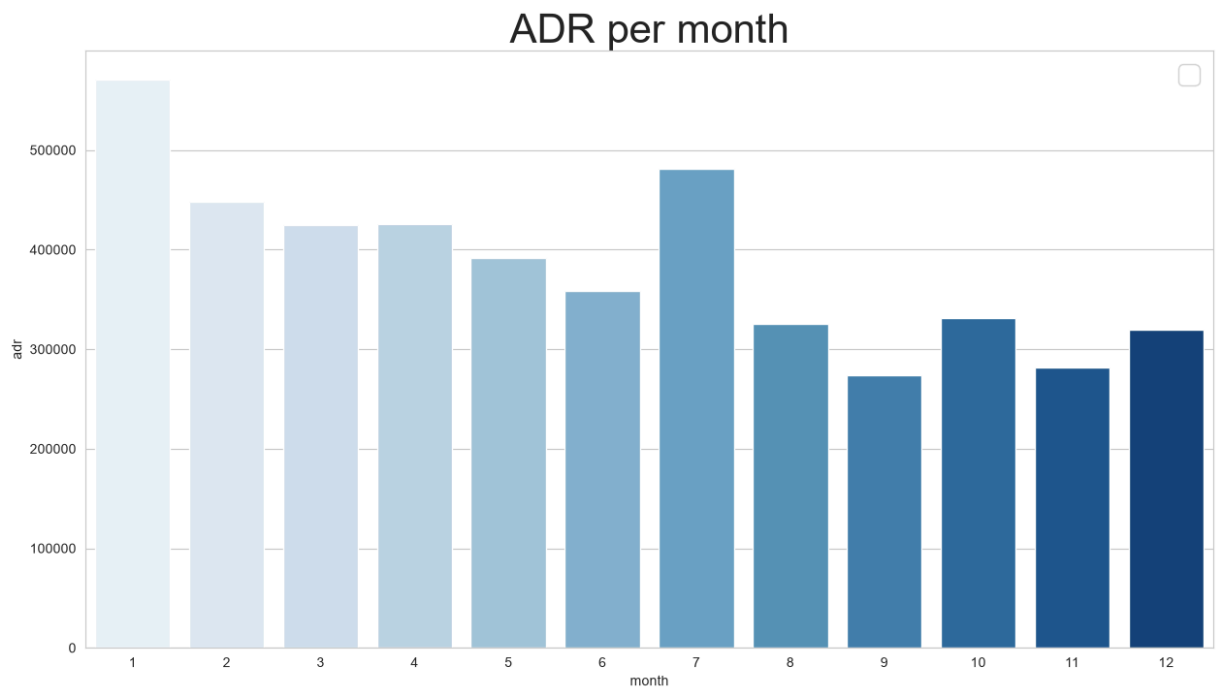


```
In [86]: df['month'] = df['reservation_status_date'].dt.month
plt.figure(figsize=(16, 8))

ax1 = sns.countplot(x='month', hue='is_canceled', data=df)
```

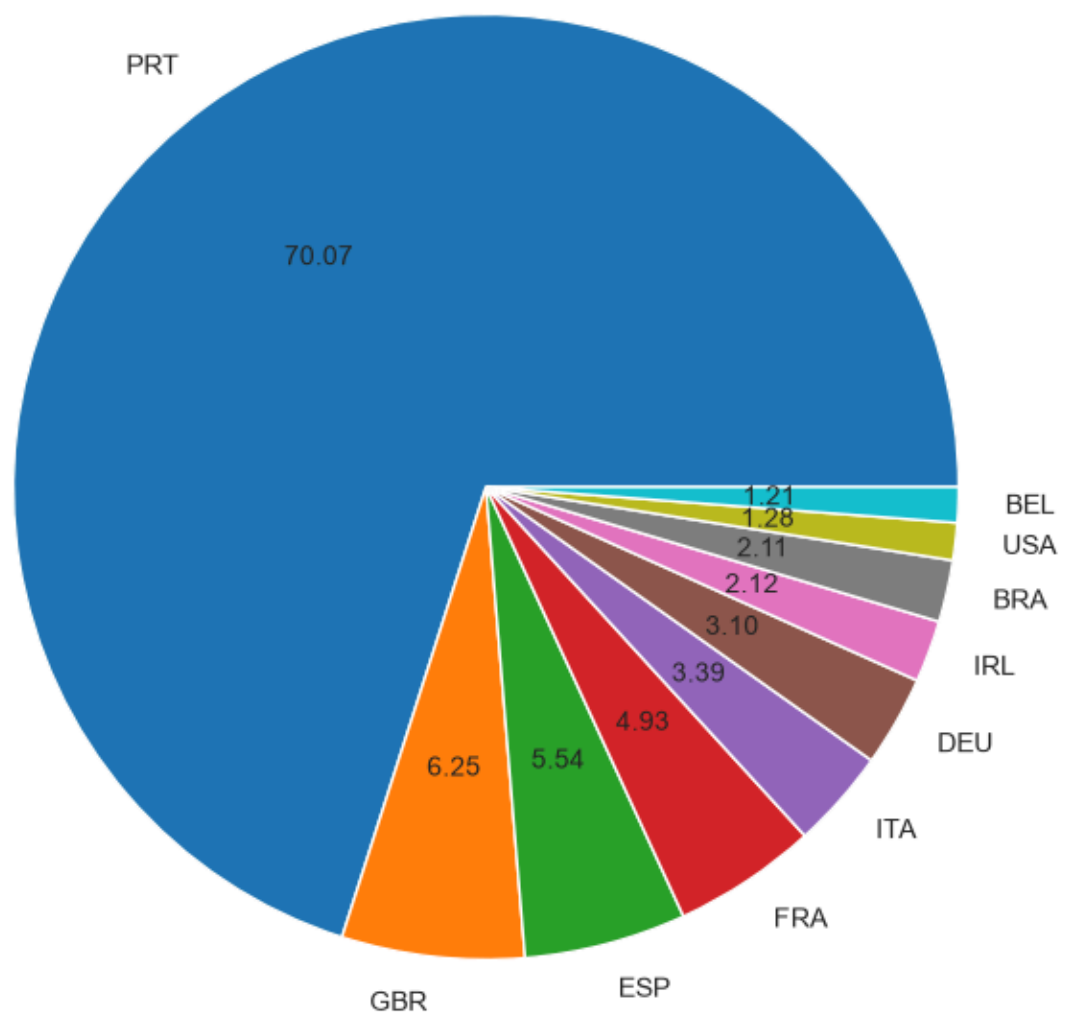


```
In [91]: plt.figure(figsize= (15,8))
plt.title("ADR per month", fontsize = 30)
sns.barplot(x = 'month', y = 'adr', data = df[df['is_canceled'] == 1].groupby('month'))
plt.legend(fontsize=20)
plt.show()
```

```
In [94]: cancelled_data = df[df['is_canceled'] == 1]
top_10_country = cancelled_data['country'].value_counts()[:10] # value_count show
plt.figure(figsize=(8,8))
plt.pie(top_10_country, autopct='%.2f', labels=top_10_country.index)
```

```
Out[94]: <matplotlib.container.PieContainer at 0x1f7b62ba990>
```



```
In [95]: df['market_segment'].value_counts()
```

```
Out[95]: market_segment
Online TA      56402
Offline TA/TO  24159
Groups         19806
Direct         12448
Corporate       5111
Complementary   734
Aviation        237
Name: count, dtype: int64
```

```
In [97]: df['market_segment'].value_counts(normalize=True) * 100
```

```
Out[97]: market_segment
Online TA      47.437698
Offline TA/TO  20.319268
Groups         16.658116
Direct         10.469566
Corporate       4.298679
Complementary   0.617341
Aviation        0.199332
Name: proportion, dtype: float64
```

```
In [98]: cancelled_data['market_segment'].value_counts(normalize=True) * 100
```

```
Out[98]: market_segment
Online TA      46.969560
Groups         27.398532
Offline TA/TO  18.746603
Direct         4.348614
Corporate       2.215075
Complementary   0.203841
Aviation        0.117775
Name: proportion, dtype: float64
```

```
In [103... cancelled_df_adr = cancelled_data.groupby('reservation_status_date')[['adr']].mean(
cancelled_df_adr.reset_index(inplace=True)
cancelled_df_adr.sort_values('reservation_status_date', inplace=True)

not_cancelled_data = df[df['is_canceled'] == 0]
not_cancelled_df_adr = not_cancelled_data.groupby('reservation_status_date')[['adr']]
not_cancelled_df_adr.reset_index(inplace=True)
not_cancelled_df_adr.sort_values('reservation_status_date', inplace=True)

plt.figure(figsize=(20,4))

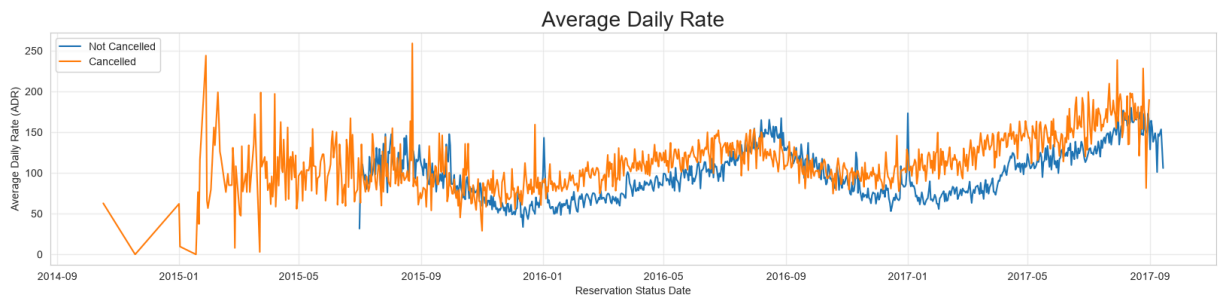
plt.title('Average Daily Rate', fontsize=20)

plt.plot(
    not_cancelled_df_adr['reservation_status_date'],
    not_cancelled_df_adr['adr'],
    label='Not Cancelled'
)

plt.plot(
    cancelled_df_adr['reservation_status_date'],
    cancelled_df_adr['adr'],
    label='Cancelled'
)

plt.xlabel("Reservation Status Date")
plt.ylabel("Average Daily Rate (ADR)")
plt.legend()
plt.grid(alpha=0.3)

plt.show()
```



```
In [104... cancelled_df_adr = cancelled_df_adr[
    (cancelled_df_adr['reservation_status_date'] > '2016') &
    (cancelled_df_adr['reservation_status_date'] < '2017-9')]

not_cancelled_df_adr = not_cancelled_df_adr[
    (not_cancelled_df_adr['reservation_status_date'] > '2016') &
    (not_cancelled_df_adr['reservation_status_date'] < '2017-9')]
```

```
In [105... plt.figure(figsize=(20,4))

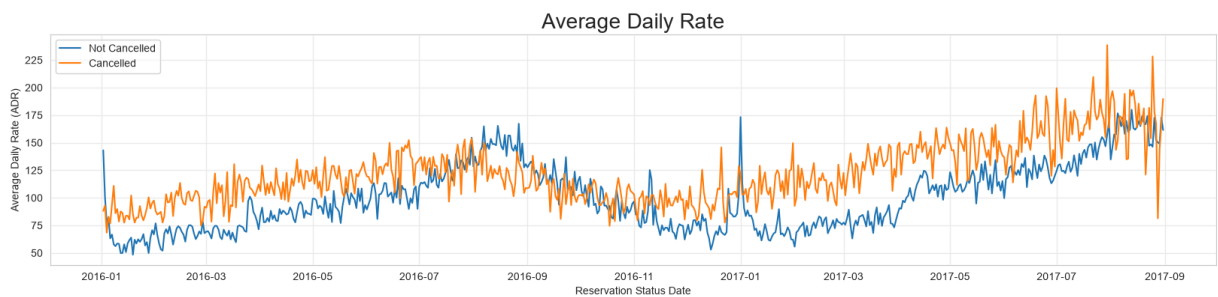
plt.title('Average Daily Rate', fontsize=20)

plt.plot(
    not_cancelled_df_adr['reservation_status_date'],
    not_cancelled_df_adr['adr'],
    label='Not Cancelled'
)

plt.plot(
    cancelled_df_adr['reservation_status_date'],
    cancelled_df_adr['adr'],
    label='Cancelled'
)

plt.xlabel("Reservation Status Date")
plt.ylabel("Average Daily Rate (ADR)")
plt.legend()
plt.grid(alpha=0.3)

plt.show()
```



```
In [ ]:
```